Day 055 | Unit 04 | Teacher plan

Numeric and Algebraic Expressions

Connect Distribution and Expression Structure

Standards: 6.EE.A.3, 6.EE.A.2c

Learning target: I can connect Distribution and Expression Structure, and explain and check my reasoning.

enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapter 5

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Multiply the factor outside parentheses by every term inside. The two expressions have the same value for every allowed input. An area model can show how one rectangle splits into smaller rectangles. Replace every variable with its assigned value, using parentheses. Apply the order of operations. In formulas, label the answer with the appropriate measurement units.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 14, 16

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01 / Learn

Multiply the factor outside parentheses by every term inside. The two expressions have the same value for every allowed input. An area model can show how one rectangle splits into smaller rectangles. Replace every variable with its assigned value, using parentheses. Apply the order of operations. In formulas, label the answer with the appropriate measurement units.

Worked example 1

Expand 6(x + 4).

1. Multiply the factor outside parentheses by every term inside.

2. The two expressions have the same value for every allowed input.

3. Multiply the outside factor by every inside term, including the constant.

4. State the result in the form requested: 6x + 24.

Answer: 6x + 24

Worked example 2

Let x = 2 and y = 5. Evaluate 4x + y². Part A: enter the value. Part B: show substitution and the order of operations.

1. Replace every variable with its assigned value, using parentheses.

2. Substitute: 4 × (2) + 5².

3. The product is 8 and the square is 25.

4. Add to get 33.

Answer: 33

Day 055 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Expand 2(x + 6).

02 / Constructed response

Evaluate 8x + 4 for x = 2.

03 / Constructed response

Expand 5(x + 2).

04 / Constructed response

Evaluate x^2 + 8y for x = 4 and y = 5.

05 / Constructed response

Expand 7(6x + 3).

06 / Constructed response

Evaluate x^2 + 2y for x = 4 and y = 2.

07 / Constructed response

Expand 7(7x + 6). Verify your result at x = 2.

08 / Numeric entry and reasoning

Let x = 1 and y = 4. Evaluate 3x + y². Part A: enter the value. Part B: show substitution and the order of operations.

09 / Multiple select

Select all expressions equivalent to 3(x + 2).
A. 6 + 3x
B. 5x
C. 3x + 6
D. 3x + 2

10 / Constructed response

A rectangle has length x + 3 and width 6. Use A = lw to write its area expression and find A when x = 5.

Day 055 | Unit 04 | Answer key

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1. 2x + 12

Multiply the outside factor by every inside term, including the constant.

6.EE.A.3 | Constructed response

2. 20

Substitute first: 8 × 2 + 4 = 20.

6.EE.A.2c | Constructed response

3. 5x + 10

Multiply the outside factor by every inside term, including the constant.

6.EE.A.3 | Constructed response

4. 56

Substitute first: 4^2 + 8 × 5 = 56.

6.EE.A.2c | Constructed response

5. 42x + 21

Multiply the outside factor by every inside term, including the constant.

6.EE.A.3 | Constructed response

6. 20

Substitute first: 4^2 + 2 × 2 = 20.

6.EE.A.2c | Constructed response

7. 49x + 42; both give 140 at x = 2

Multiply the outside factor by every inside term, including the constant.

6.EE.A.3 | Constructed response

8. 19

Substitute: 3 × (1) + 4². The product is 3 and the square is 16. Add to get 19.

6.EE.A.2c | Numeric entry and reasoning

9. A, C - 6 + 3x; 3x + 6

Distribute 3 to both terms to get 3x + 6. Reversing the addends gives the same value. The other expressions fail, for example at x = 0.

6.EE.A.3 | Multiple select

10. A = 6(x + 3); 48 square units

Substitute x = 5: 6(5 + 3) = 48.

6.EE.A.2c | Constructed response